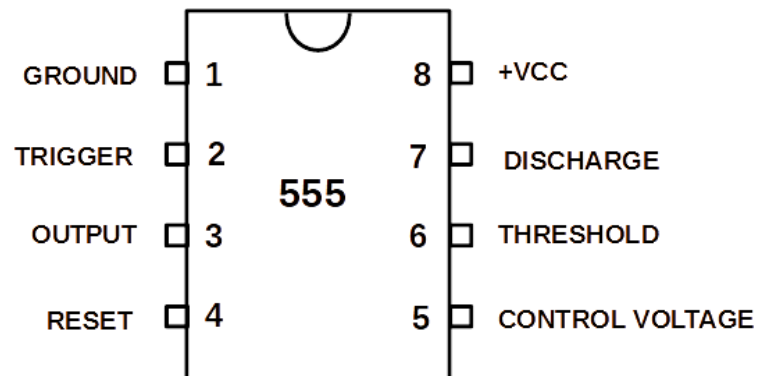


ASTABLE AND MONOSTABLE MULTIVIBRATORS USING 555 TIMER

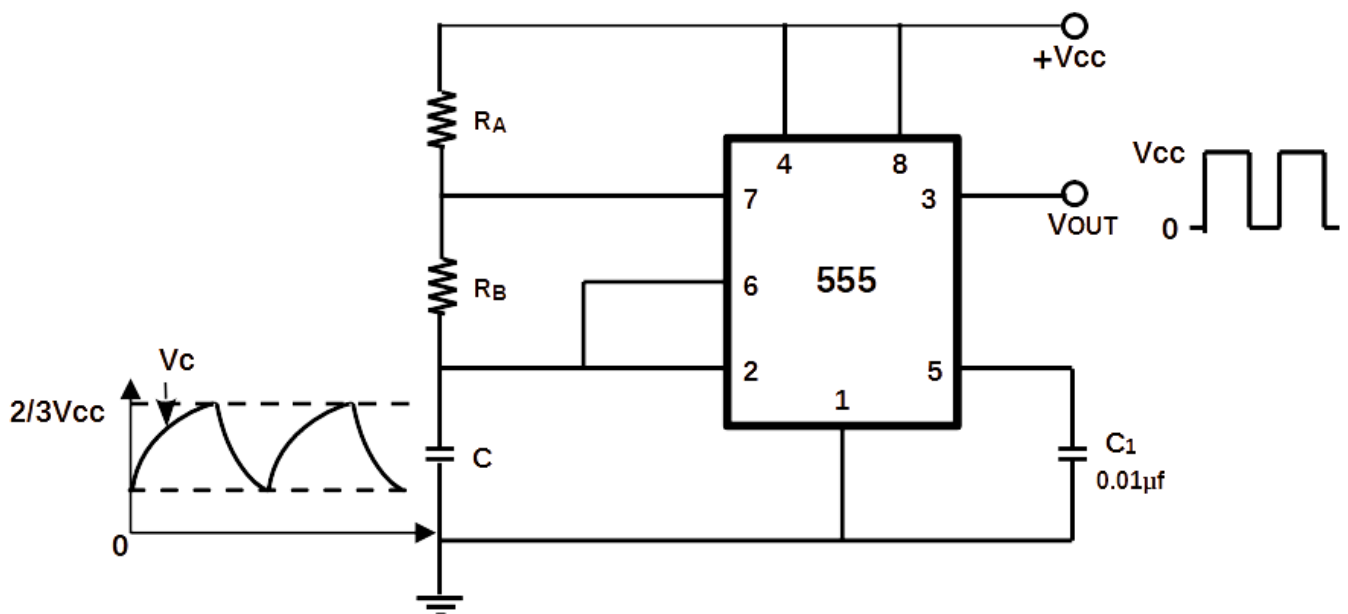
AIM:

To design and set up an astable multivibrator and a monostable multivibrator using 555 timer.

PIN Diagram of IC 555



Astable Multivibrator



Design:

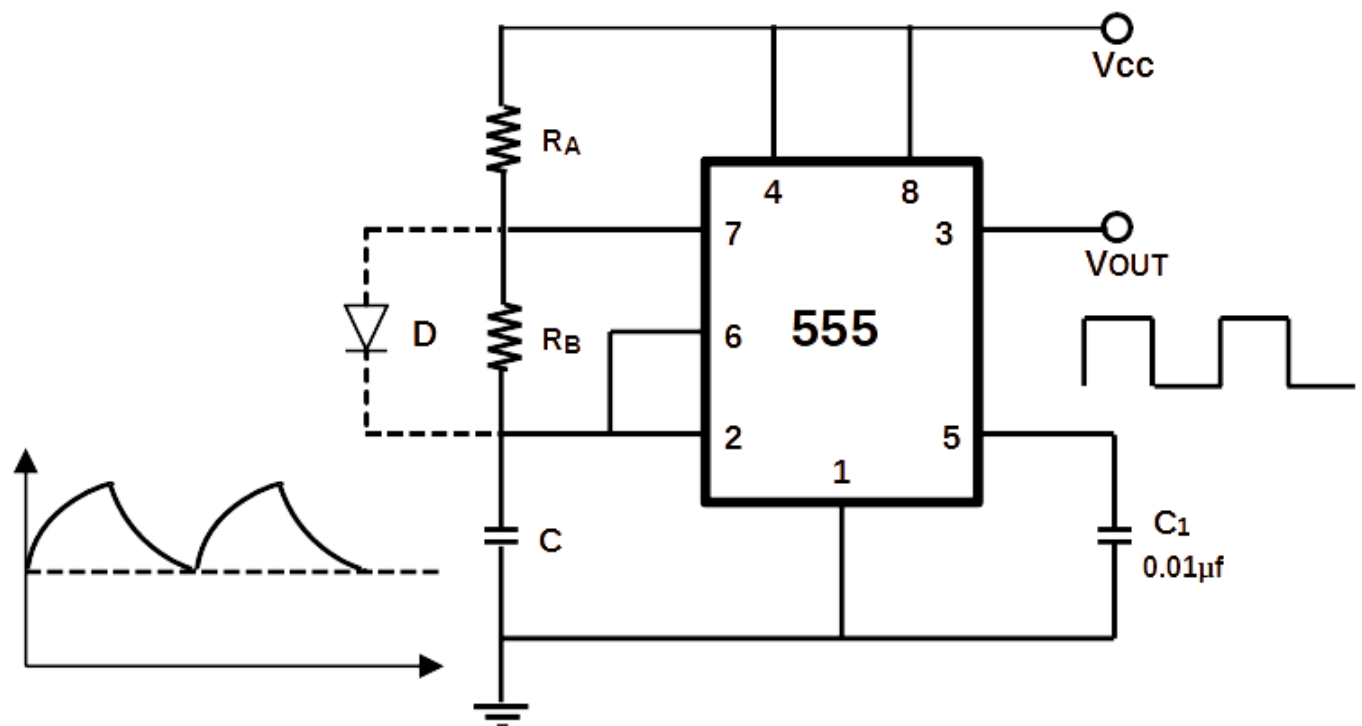
Charging time $t_c = 0.69(R_A + R_B) C = 1\text{msec}$

Discharging time $t_d = 0.69 R_B C = 0.5\text{msec}$

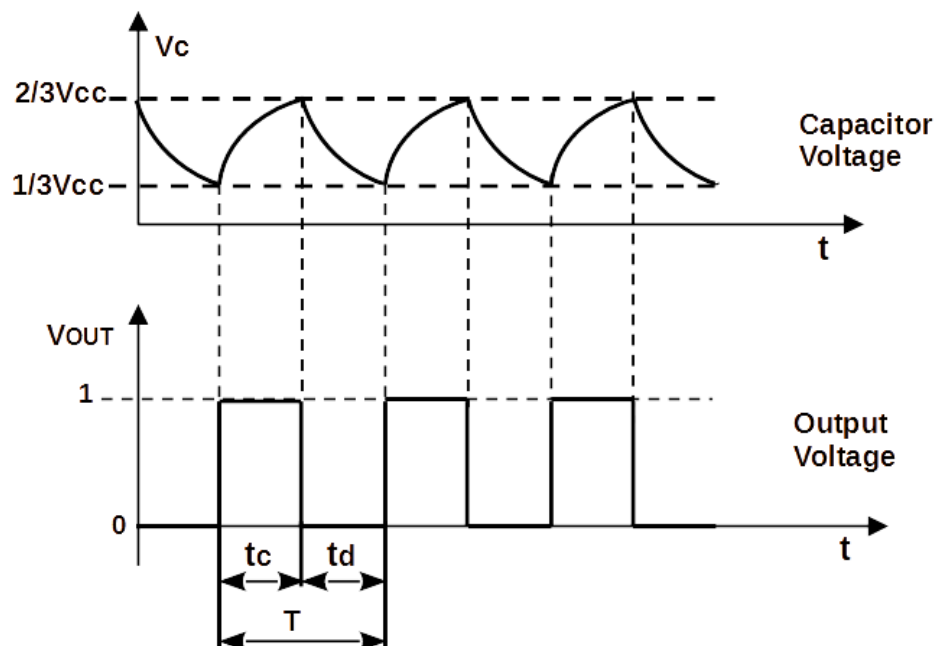
Let $C = 0.1\mu\text{F}$, then $R_B = 6.8\text{K}\Omega$

Choose $C_1 = 0.01\mu\text{F}$, $R_A = R_B = 6.8\text{K}\Omega$

Astable with 50% Duty cycle



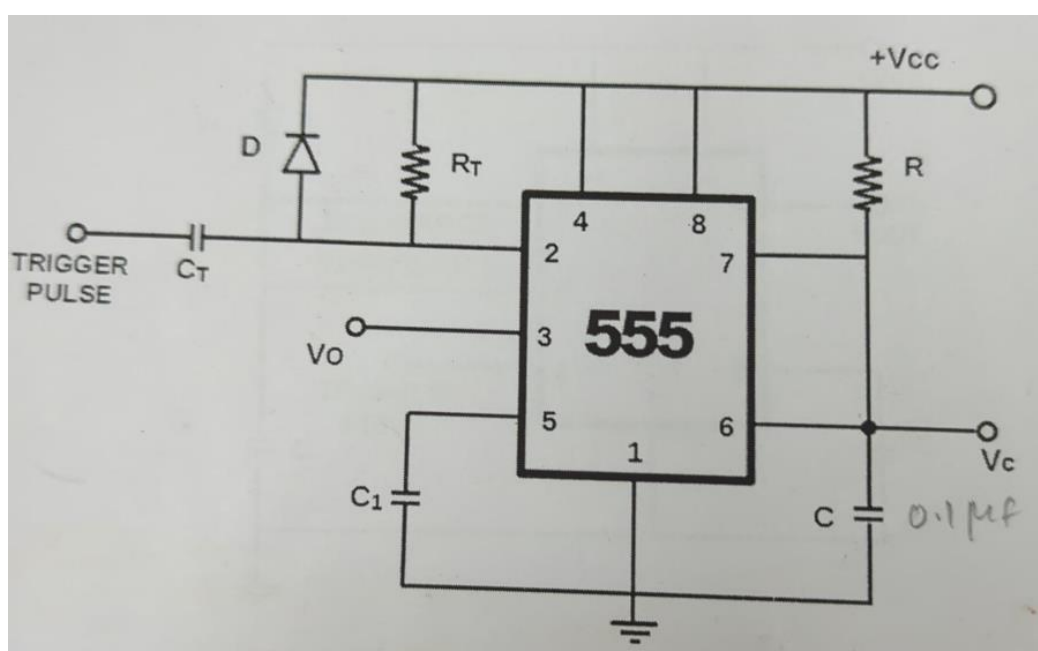
Output Waveforms

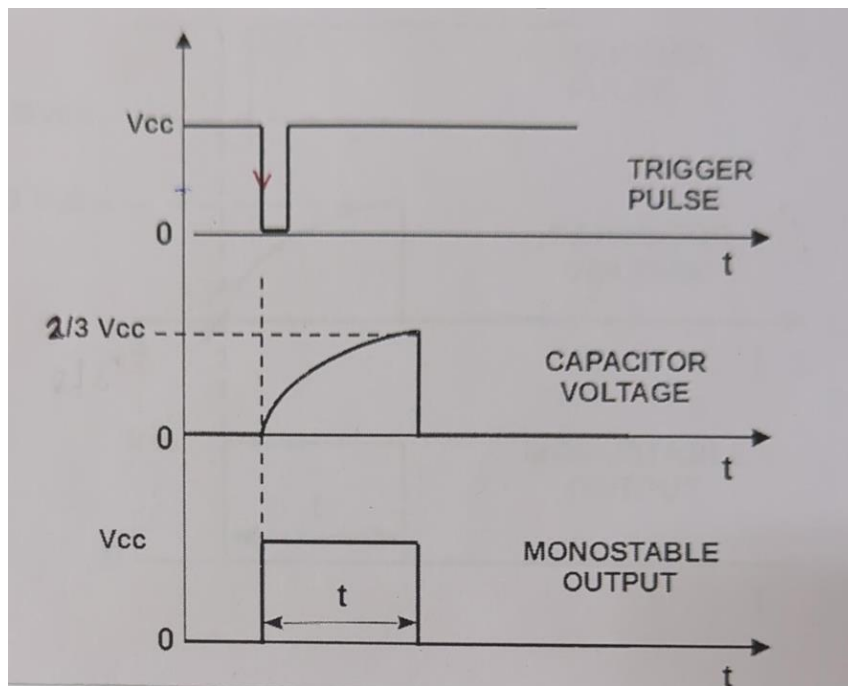
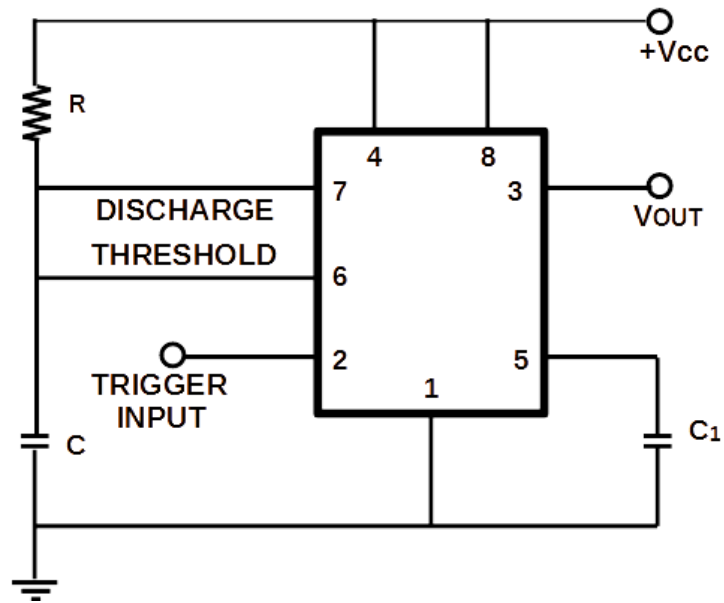


MONOSTABLE MULTIVIBRATOR

Aim:

To Design and set up a monostable multivibrator using 555 timer for a pulse width of 1ms.





Design

Pulse width $T = 1.1RC = 1\text{ms}$

Let $C = 0.1\text{ }\mu\text{F}$, then $R = 10\text{K}\Omega$

Design of the triggering circuit:

$RTCT \leq 0.0016 T_t$ where T_t is the time period of the trigger.

Let $T_t = 3\text{ms}$ and $C_T = 0.01\text{ }\mu\text{F}$, then $R_T = 480\Omega$

Choose $C_1 = 0.01\text{ }\mu\text{F}$

RESULT: